

Project Blueprint: Insurance Renewal Management Portal

A Comprehensive Solution for Multi-Company Insurance Agents

1. Executive Summary

This project aims to develop a centralized internal portal for insurance agents to manage policy renewals across multiple providers (HDFC Life, Niva Bupa, ICICI Lombard, etc.). The system will automate the tracking of premium due dates, send multi-channel alerts (SMS/Email/WhatsApp), and provide an internal dashboard for agents to monitor payment statuses and client interactions.

2. Core Functional Requirements

2.1 Automated Alert System

- **Advance Notification:** Automatically send renewal reminders 30 days before the due date to both the client and the internal agency team.
- **Delinquency Reminders:** If a policy status remains 'Unpaid' after the due date, the system triggers bi-weekly (twice a week) follow-up alerts until updated.
- **Multi-Channel Delivery:** Alerts via Email and SMS/WhatsApp API integration.

2.2 Internal Renewal Tracking & CRM

- **Payment Status:** Toggle switch or dropdown for each policy (Paid / Pending / Grace Period / Lapsed).

- **Interaction Logs:** Add remarks for each client with automatic Date and Time stamping (e.g., "Client promised to pay by Friday").
- **Chronological View:** Dashboard default view sorted by "Upcoming Due Date" to ensure priority focus.

2.3 Client Onboarding & Data Management

- **Bulk Import:** Upload existing client data via CSV/Excel.
- **Manual Entry:** Form to add new clients with fields: Name, Company, Premium, Policy No, Contact Details, and Issuance Date.
- **Search & Filter:** Robust filters by Insurance Company name, Date range (due dates), and Client name/Phone.

3. Technical Data Schema

Field Name	Data Type	Description
Client Name	String	Full name of the policyholder.
Insurance Company	Dropdown	HDFC Life, ICICI Lombard, Niva Bupa, etc.
Policy Number	Alphanumeric	Unique identifier for the policy.
Premium Amount	Decimal	The amount due for renewal.
Due Date	Date	The deadline for premium payment.
Issuance Date	Date	Date the policy was first issued.
Contact Info	Phone / Email	Target for automated alerts.
Status	Enum	Paid, Pending, Overdue.

4. Workflow Logic

1. **Data Entry:** New policy is added or imported from the agent's master file.
2. **The T-30 Trigger:** 30 days before "Due Date", the server sends the first notification to the client and adds a task to the agent's dashboard.
3. **The Renewal Window:** The agent calls the client; any conversation is logged in the "Remarks" section with a timestamp.
4. **Payment Update:** If "Paid", the agent updates the status, stopping further alerts and calculating the next year's due date.
5. **Overdue Protocol:** If "Pending" past due date, bi-weekly alerts are sent automatically.

5. Proposed Tech Stack

- **Frontend:** Next.js (for a responsive, fast dashboard).
- **Backend:** Node.js (Express)
- **Database:** Supabase (to handle structured relational data).
- **Automation:** Cron Jobs (to check due dates daily and trigger emails/SMS).
- **Communications:** Twilio (SMS/WhatsApp) and SendGrid/AWS SES (Email).